

OSCEOLA, WI, USA

WMO: 726418

Lat: 45.308N

Lon: 92.690W

Elev: 275

StdP: 98.06

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 54912

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-25.9	-22.7	-29.3	0.3	-24.3	-27.2	0.3	-22.2	9.4	-6.9	8.6	-7.2	1.2	290	0.551

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.4	31.4	23.2	29.8	22.4	28.0	21.1	25.3	29.4	23.9	27.9	22.8	26.6	4.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.5	27.6	22.6	18.0	26.1	21.7	16.9	25.2	78.7	29.3	73.1	28.2	68.6	26.7	32.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 9.2	(b) 8.2	(c) 7.3	(e) -29.8	(f) 34.4	(g) 3.0	(h) 1.8	(i) -32.0	(j) 35.6	(k) -33.7	(l) 36.7	(m) -35.4	(n) 37.7	(o) -37.7	(p) 38.9		
(a) 9.2	(b) 8.2	(c) 7.3	(e) -30.0	(f) 26.8	(g) 3.0	(h) 1.5	(i) -32.2	(j) 27.9	(k) -33.9	(l) 28.8	(m) -35.6	(n) 29.6	(o) -37.8	(p) 30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.0	-9.4	-8.1	-0.9	7.3	14.0	19.3	21.7	19.8	16.1	8.5	1.2	-6.6		
	DBStd	12.23	7.13	6.90	6.62	5.56	4.76	3.65	3.03	3.32	4.70	5.07	5.77	6.76		
	HDD10.0	2460	603	506	342	114	15	0	0	1	6	90	268	516		
	HDD18.3	4430	861	739	597	332	150	31	7	21	98	307	513	774		
	CDD10.0	1360	0	0	3	34	139	279	362	305	188	44	5	0		
	CDD18.3	287	0	0	0	2	16	60	110	67	29	3	0	0		
	CDH23.3	2630	0	0	1	25	186	543	1026	582	246	23	0	0		
	CDH26.7	736	0	0	0	6	55	151	312	152	58	2	0	0		
(14) Wind	WSAvg	3.0	3.0	3.1	3.2	3.7	3.5	3.0	2.4	2.2	2.8	3.1	3.2	3.1		
(15) Precipitation	PrecAvg	834	23	21	42	72	100	118	109	102	94	79	45	27		
	PrecMax	1107	56	46	82	205	198	195	198	215	219	181	120	63		
	PrecMin	605	4	2	15	17	31	47	55	25	29	26	1	6		
	PrecStd	134	14	11	19	40	48	37	43	49	47	47	35	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.0	10.4	18.9	27.1	32.0	32.8	33.7	32.5	31.1	26.3	18.9	8.3		
		MCWB	3.3	7.3	12.1	17.1	20.0	24.1	25.1	24.0	22.3	17.0	13.1	6.7		
	2%	DB	3.5	6.3	14.6	22.5	27.7	30.2	31.7	30.1	28.1	22.7	15.9	6.0		
		MCWB	1.7	3.6	10.1	13.1	18.7	21.5	24.3	23.1	21.1	16.1	11.1	3.9		
	5%	DB	2.1	3.6	11.2	19.7	25.1	27.9	29.9	28.0	26.3	19.8	12.4	3.1		
		MCWB	1.1	1.5	8.0	11.5	16.8	20.5	23.1	21.6	19.8	14.5	8.6	1.8		
	10%	DB	0.2	2.0	7.8	17.3	22.6	26.8	28.0	27.0	24.0	17.2	9.8	2.2		
		MCWB	-0.6	0.4	5.0	10.5	15.5	19.8	21.7	20.9	18.7	12.8	6.9	1.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	7.4	13.8	18.0	22.6	27.0	28.0	26.2	24.4	20.5	14.8	7.8		
		MCDB	6.0	10.5	16.4	24.5	29.1	29.3	30.5	29.7	27.9	23.0	17.6	8.0		
	2%	WB	2.0	3.8	10.6	15.0	20.4	24.1	25.7	24.3	22.8	17.1	11.7	4.0		
		MCDB	3.2	5.9	13.8	20.5	24.9	27.9	29.6	28.2	26.4	20.4	15.0	5.3		
	5%	WB	0.9	1.8	7.8	12.7	18.8	22.4	24.2	23.0	21.3	15.2	8.9	2.4		
		MCDB	1.9	3.4	10.7	17.8	22.7	26.3	28.4	26.5	24.8	19.1	11.5	3.3		
	10%	WB	-0.6	0.2	5.4	11.1	17.3	20.9	23.0	21.8	19.8	13.1	7.0	1.2		
		MCDB	0.2	1.6	7.9	16.3	20.7	24.5	26.9	25.2	23.3	16.6	9.6	2.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	10.2	10.0	11.2	11.9	11.0	11.4	11.5	11.7	10.5	8.8	8.0		
		MCDBR	9.3	10.4	13.1	15.9	15.2	12.9	12.9	13.0	13.2	14.3	12.6	8.7		
		MCWBR	7.7	8.1	8.5	8.7	7.3	6.3	6.5	6.6	6.9	8.3	8.4	7.4		
		MCDBR	8.9	9.6	12.3	13.7	12.0	10.8	10.8	11.1	11.0	11.5	11.1	7.7		
	5% WB	MCWBR	7.6	7.8	8.5	8.5	6.9	6.8	6.4	6.7	6.7	7.7	8.1	6.7		
(40) Clear-Sky Solar Irradiance	taub		0.259	0.279	0.319	0.371	0.403	0.392	0.385	0.397	0.368	0.319	0.288	0.265		
	taud		2.401	2.337	2.348	2.303	2.259	2.328	2.373	2.348	2.401	2.526	2.509	2.421		
	Ebn at Noon		867	910	913	888	866	875	877	852	849	856	827	819		
	Edh at Noon		71	92	106	122	132	124	117	116	101	77	65	63		
(44) All-Sky Solar Radiation	RadAvg		1.71	2.80	3.87	4.62	5.33	5.91	6.25	5.21	4.05	2.54	1.61	1.19		
	RadStd		0.19	0.27	0.34	0.45	0.43	0.45	0.21	0.38	0.28	0.34	0.14	0.15		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	-1.03	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (58 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page